ASTR 302: Scientific Papers, Writing, and References



Agenda – November 2nd , 2026

- **Mock Journal Club/Reading Day – November 4th , 2026**
  BRING LAPTOP!!!!!!!!!!!!

- **Observing Check in and Discussion**

- **Structure of a Scientific Paper**

Summary of Paper Choice Due November 13th, 2026

Agenda – November 2nd , 2026

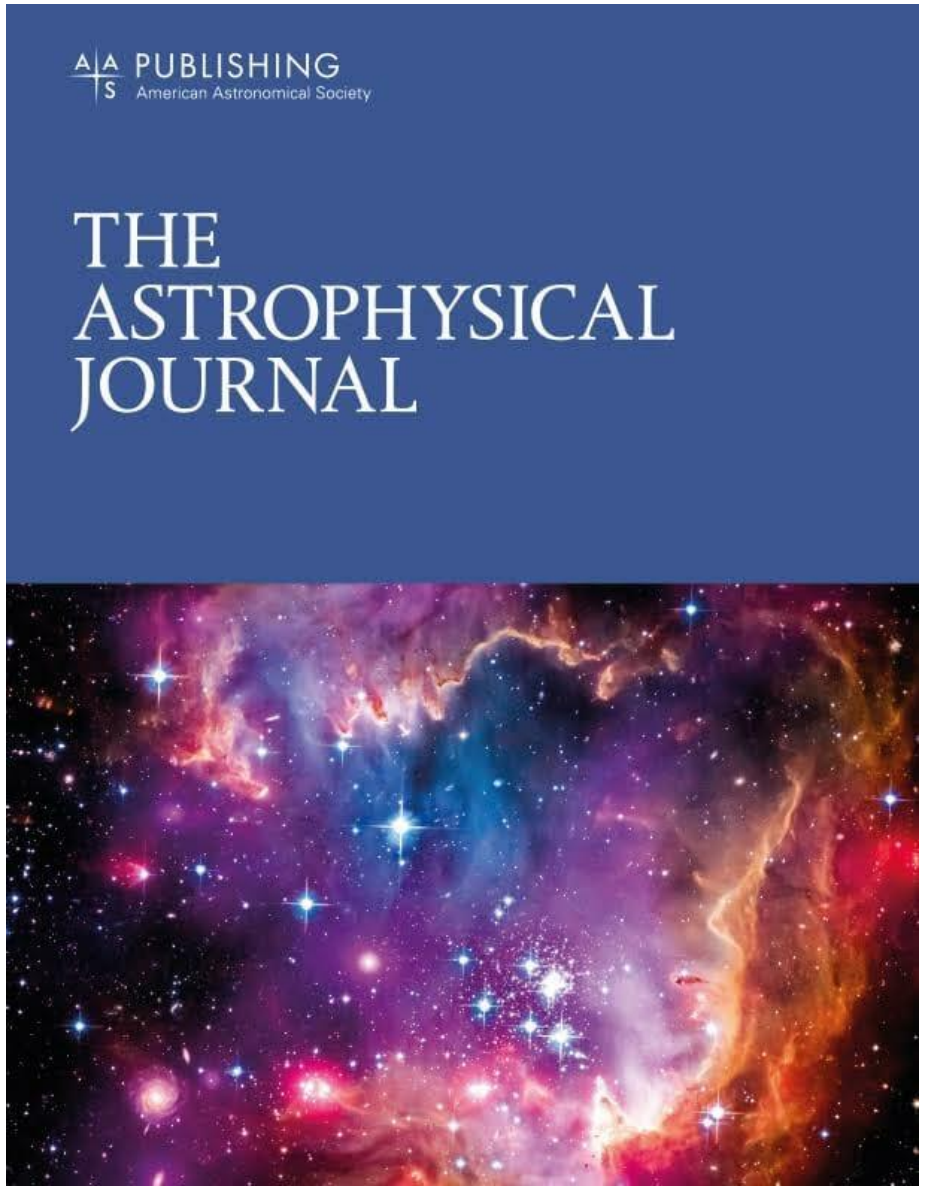
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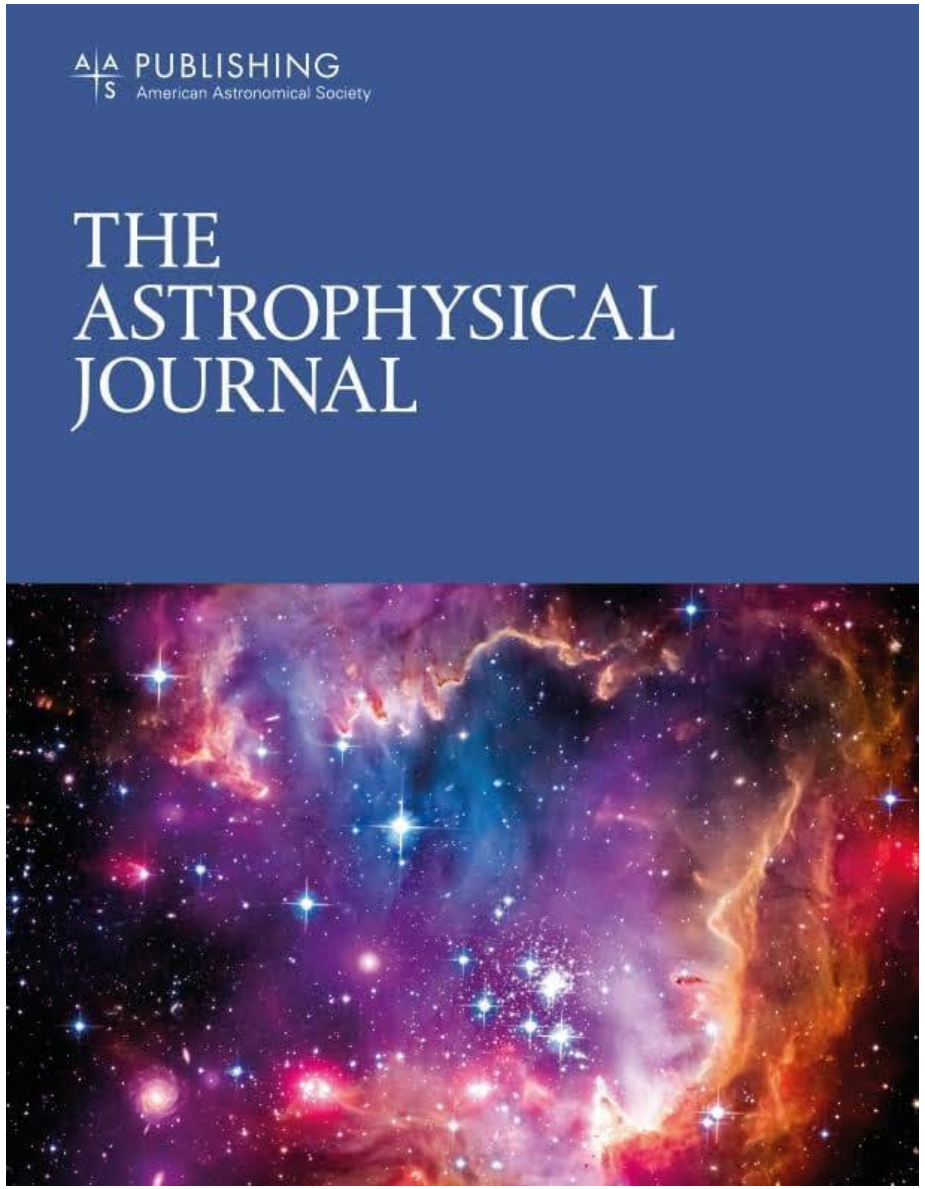
  - "The Discovery of a Very Cool, Very Nearby Brown Dwarf in the Galactic plane" – Lucas+ 2010 **(Imaging + Spectroscopy)**
  - "The Discovery of a Luminous Z=5.80 Quasar from the Sloan Digital Sky Survey" – Fan+ 2000 **(Imaging + Spectroscopy)**
  - "High Proper-Motion Stars in the Vicinity of Sagittarius A*: Evidence for a Supermassive Black Hole at the Center of Our Galaxy" – Ghez+ 1998 **(Imaging)**

Summary of Paper Choice Due November 13th, 2026

Why Do We Do Scientific Writing?

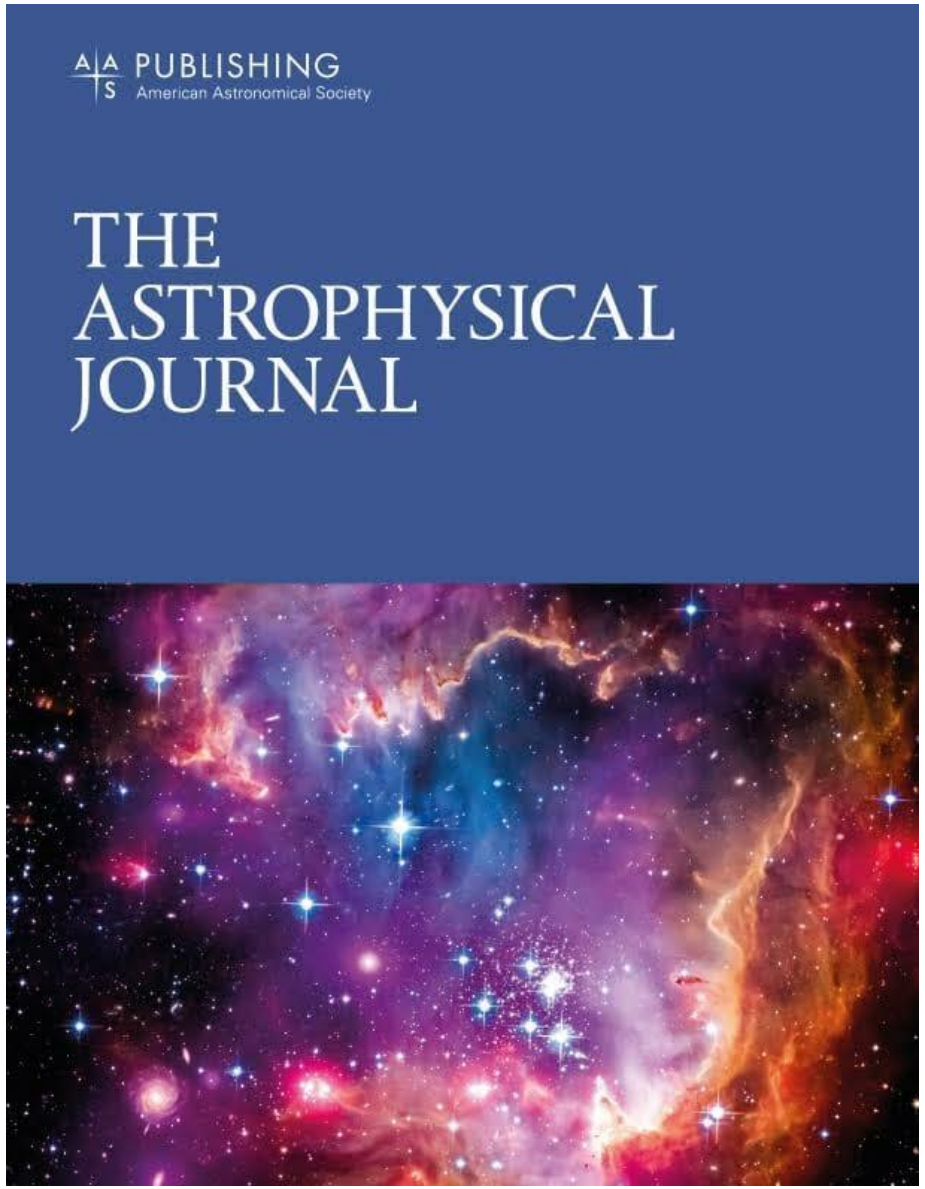


Why Do We Do Scientific Writing?



- To share results/ideas with other scientific peers clearly and concisely.
- Writing forces you to think and helps you create new ideas.
- Learning how to develop/motivate ideas with logic makes you a more persuasive writer.

Why Do We Do Scientific Writing?



- You need to write to build a research group (pay people) and academic reputation.
- Write documentation to make other people's lives easier. Robust science is reproducible.
- Self expression

General Structure of Research Paper

- Abstract (Short Summary)
- Introduction
- Methods/Observations
- Results
- Discussion/Summary
- Acknowledgements
- References

General Structure of Research Paper

- **Introduction**
  - Provide context to your reader on the question you want to answer, past work, and why your work is important. Describe the outline of your paper.

- **Methods/Observations**
  - Describe the telescope/instrument mode(s) used and what date(s) you observed.
  - Describe sky/atmospheric conditions (seeing, water vapor, moon, logs).
  - Document how many images were taken at what integration time. How much total time you spent on each target.
  - Describe the steps you did to reduce your astronomical images.
  - This section can often be split into three parts: **Sample, Observations, Data Analysis.**

General Structure of Research Paper

- **Results**
  - This is where you talk about your derived quantities from your data
    - Brightness of targets (photometry)
    - Abundances/Spectral features (spectroscopy)
    - Periodicity (time series photometry or spectroscopy)
    - Radial velocity (spectroscopy)
    - etc

- **Discussion**
  - Why are your derived quantities important to the scientific question you want to answer?
  - What is the new information you provided to the field?
  - How does your work compare to others?
  - What work needs to be done in the future?

General Structure of Research Paper

- **Acknowledgements**
  - Who funded your research. (Funded by NSF grant #2344a-xDD)
  - What tools/facilities did you use.
  - Land acknowledgements.
  - Observatory staff who took your data

- **References**
  - If an idea or background concept is not your own you must cite it.
  - You typically cannot reproduce figures from someone else without asking first.
  - You can cite software for python routines you used.

Planning Your Research Paper

- **Think about what message you want to convey to your readers**
  - ✅ My new observations tell us X new thing about this object
  - ❌ I did data reduction, look at my pictures/spectra

- **Revisit your proposal you submitted early in the semester. Keep asking yourself why you even did this project.**

- **Storyboard your Sections**
  - Write out what ideas/topics you want to touch on in each section
  - Make pictures of what figures you want/need.

References

- **Refereed vs Non-Refereed Papers**
  - If a paper was refereed it has gone through some peer review process where someone not on the paper provided feedback or edits.

  - **Reviewed and Non-reviewed papers end up on arxiv.org so always try to look for a direct journal link when possible.**

- **Resources**
  - **Astrophysics Data System (https://ui.adsabs.harvard.edu/)**
  - **Science Explorer (https://scixplorer.org/)**
  - **“The arxiv” Arxiv.org**
  - **SIMBAD (https://simbad.cds.unistra.fr/simbad/sim-fbasic)**

DEMO